Divine Proportions: Exploring Sacred Geometry in Safavid Domes (Case Study: Sheikh Lotfollah Mosque)

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Abstract

This paper explores the impact of sacred art and sacred geometry on the philosophies of the East and West. It also looks at how these perspectives influenced the development of sacred art in Iran, focusing on the Persian-Islamic art and architecture in the Safavid period. The research unfolds with a dual purpose using a descriptive and analytical approach. First, it seeks to unravel the deep-rooted relationship between sacred geometry and Persian-Islamic geometry. Drawing insights from Neoplatonism, the Peripatetic school, and the Theory of Forms, the study explores the historical discourse on divine art and its influence on Persian-Islamic artistic expressions. Case studies are employed to illuminate connections between these philosophical frameworks and the geometry seen in Safavid domes.

Subsequently, the research meticulously analyzes the constituent patterns of Islamic geometric designs during the Safavid era. Through case studies and examinations of architecture, particularly focusing on the renowned Sheikh Lotfollah Mosque, the study investigates the intricate interplay between sacred geometry and the geometric patterns embedded in the construction of domes and associated tile work.

Ultimately, Persian-Islamic art and architecture, steeped in the Philosophy of Illumination and Plato's theory of forms, are unveiled as repositories of intricate geometric patterns. These patterns, rooted in divine philosophy, hold symbolic significance in our subconscious, offering a pathway to divine perceptions and a profound connection between art, philosophy, and the divine meaning of life.

Keywords: Persian-Islamic Art, Sacred Geometry, Islamic Geometry, Safavid Dynasty, Platonic Solids, Sufism, Illuminationism, Seyyed Hossein Nasr

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Introduction

Within the realm of visual arts, we come to understand that our very being comprises symbolic representations. Each concept, tangible object, definition, perception, and visual correlation holds meaningful significance that we encounter time and time again throughout our existence. These symbols influence our journey in life and provide us with a deeper insight and comprehension of the world that surrounds us. Throughout history, individuals seeking enlightenment have pursued the elusive concept of the meaning of life. They embark upon this quest at various junctures in their lives, relying on intuition and exploration to guide them. Artists, in particular, are known for their role as seekers, using their creative talents and reflective observations to derive deeper understandings or to position themselves within the present moment during the art-making process. The experience of wonder during the creation and observation of a work is an experience that imparts meaning to art. It can be considered a spiritual experience in which the artist's awareness undergoes transformation.

Foundation geometry and Islamic geometry share a common root. For example, light is the source of geometry in the realm of Illuminationism in Iran and the World of Archetypes in Plato's Theory of Forms. Then, forms and complex patterns emerge from geometry, each reflecting a specific meaning in philosophical schools.

"Sacred geometry in Islamic art also encompasses numbers and proportions that lead towards unity through various divisions and forms." (Kamrani, 2000, p. 114)

Many separate studies have been conducted on sacred geometry and Islamic geometry, but a question that consistently engages the mind while studying these researches is their interrelation. Most studies focus on examining Islamic geometric patterns in different periods and attempt to

categorize them. However, It is rare to find research that can trace patterns back to fundamental geometry, namely sacred geometry, descriptively and analytically.

By analyzing the geometric patterns, especially miniature patterns in the tile works of domes and vaults, we can reach abstract forms of sacred geometry that serve as the foundation for shaping these patterns. These patterns have evolved more comprehensively over time based on specific logic and function throughout history. With an understanding of these functions, we can fill the mentioned void and discover patterns that can be claimed as fundamental to Islamic geometric patterns in the specified period. Access to these patterns provides the necessary information for creating modern artistic works with a sacred artistic essence.

Research Background

Among the domestic research conducted in the field of examining and introducing sacred geometry in the Persian language, Hejazi's research (2005), "Sacred Geometry in Nature and Persian Architecture," stands out. This research thoroughly investigates the roots of some forms and patterns used in Persian architecture during the Islamic period and their connection with mathematical proportions in nature. Another noteworthy study is Lotfi's thesis (2010), "نظام‌های هندسی و طراحی نقوش گردان در ایران [Geometric Systems and Design of Rotational Patterns in Iran]" which extensively examines many recurring patterns in Islamic designs, particularly in the tile works of domes. Moreover, research has been conducted on the philosophy of these geometric patterns, miniature illustrations, and the concept of sacred art. Notable among these studies are Asadi's thesis (2016), " مبانی هندسه مقدس در فلسفه اسلامی [Investigating the Theoretical Foundations of Sacred Geometry in Islamic Philosophy]" and Hajari's thesis (2011), " مبانی هنر قدسی در فلسفه شرق (ایران–صفویان) [Foundations of Sacred Art in Eastern Philosophy (Iran – Safavids)]" These studies delve into the intellectual foundations behind Islamic art and geometry and their connection to sacred art.

In the realm of Islamic geometric patterns, there is a noticeable void that could benefit from a fresh approach and the utilization of modern tools to engage in the production of contemporary artistic works in this field. One avenue for such exploration is the integration of a novel perspective and the application of modern tools to create art relevant to today's media. Additionally, in international research, studies have delved into the examination and simplification of the constitutive patterns of Islamic geometry. Notable among these are Laura U. Marks's book "Enfoldment and Infinity: An Islamic Genealogy of New Media Art" and her Article "Infinity and Accident: Strategies of Enfoldment in Islamic Art and Computer Art", and Dabbour's (2012)

article "The Underlying Structure of Design Process for Islamic Geometric Patterns." Also, Scholars such as Robert Lawlor and Henri Corbin have contributed significantly, providing abundant theoretical and semiotic insights in this domain. The works of Stephen Skinner stand as a comprehensive reference for investigating fundamental geometric patterns.

Methodology

This research can be deemed practical in its objectives. By analyzing and describing the various roles and representations, one can uncover the distinct visual characteristics that artists have extracted from this concept. Geometric patterns in Iran, due to spatial and temporal distances, have lagged in certain aspects behind modern art. As contemporary artists seek to find their own personal style, the expressive possibilities identified in this project can serve as a suitable response to this aspiration for painters and students.

In terms of research methodology, this thesis is descriptive-analytical. This is because the research not only addresses the issue of Islamic geometric patterns but also explains how this potential influence occurs. It then presents documents related to the abstract geometric patterns of sacred geometry in the architecture of domes and the design of tile patterns. Furthermore, this research seeks to explore, from a philosophical perspective, the meaningful relationship of these geometric patterns in the human subconscious and how they are represented in modern paintings and illustrations. In the descriptive-analytic method, the researcher not only describes the issue and its dimensions but also explains its status and underlying reasons.

Theoretical concepts

It is necessary to review some basics of physics and geometry to better understand this geometry and the vibrational pattern. Examining this foundational information not only prepares the mind to comprehend more complex geometric patterns but also helps us grasp the physical dimensions or, philosophically speaking, the "intelligible world."

Bragdon[1], an architect and author, has utilized various aspects of Hinton[2]'s achievements in his theory of three-dimensional beyond. In his book "Foundations of the Fourth Dimension," he presents different geometric and philosophical methods for better understanding. (Zandi, 2017, p. 66) One of the best methods he introduces is the extension of one dimension to reach higher dimensions. As observed in the explanations and figures, this process is an evolutionary path similar to nature itself. Higher and more complex dimensions theoretically have evolved from lower and easier-to-understand dimensions for us humans. The intertwining and coexistence of these dimensions create a multi-dimensional space where forms take shape, which may initially seem challenging to us but are inherently familiar to the human subconscious.

Nature consistently manifests itself in various geometric patterns. These formative patterns, which we experience through our five senses, are continuously evolving recursively. Alongside this evolutionary path, these patterns, with their fractal property, are consistently repeating and reminding us of fundamental geometry in our subconscious. To explore the depths

1 - Claude Fayette Bragdon, American architect and writer
2 - Charles Howard Hinton, British. Founder of the word “Tesseract”

of this fundamental geometry, it is best to start with the basic foundations of spacetime[3] geometry, and that begins with a point.

Point: A single point is defined in coordinate systems like the Cartesian coordinate system. A point is the first element in visual sciences, but a point never has external existence in our physical world; for example, we can consider the number "zero" in mathematics. (Critchlow, 1969, p. 5; Ching, 1979, p. 20)

Line: The connection of two points in a coordinate system represents a vector, which we call a line. In visual aesthetics, a line is the second element, but like a point in our physical world, it has no external existence. (Critchlow, 1969, p. 5; Ching, 1979, p. 24)

Plane: By connecting three or more points in sequence through lines, a two-dimensional space is created, which we call a plane. A plane has more visual power than a point or a line, but it still lacks a physical presence in our world. (Critchlow, 1969, p. 5; Ching, 1979, p. 30)

Volume: The volume arises from connecting at least four surfaces (in order and in a manner that creates an enclosed space). A four-faced pyramid is recognized as the simplest three-dimensional volume, but here, we use a cube as an example. (Critchlow, 1969, p. 5; Ching, 1979, p. 44)

Tesseract (Four-Dimensional Cube): To visualize four-dimensional space (beyond three dimensions), we consider a three-dimensional cube with six faces on each side. The tesseract, a polytope formed by intertwining cubes, is created by replacing each face with another three-dimensional cube. The variables in a tesseract are: *(longitudinal vector, transversal vector in the*

3 - spacetime | any mathematical model which uses three dimensions of space and one dimension of time.

height vector = three-dimensional volume in the time vector) Since the physical world is three-dimensional, the only way to depict a four-dimensional polytope is to use the time factor. (Critchlow, 1969, p. 25; Convey, 2008, p. 409)

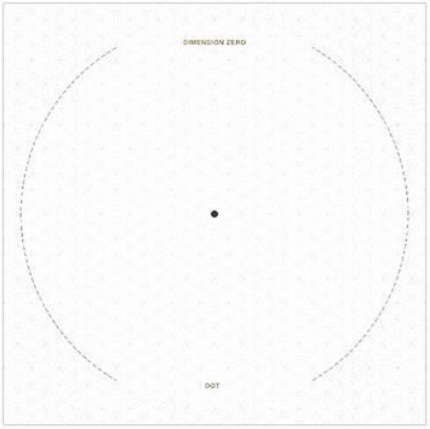
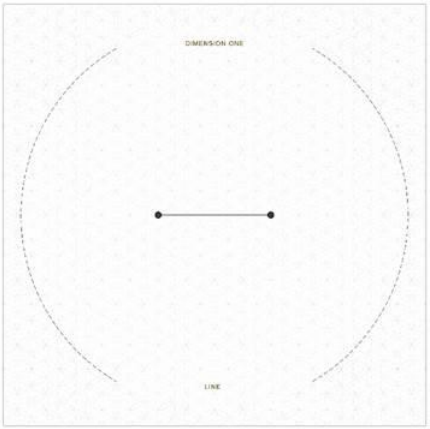
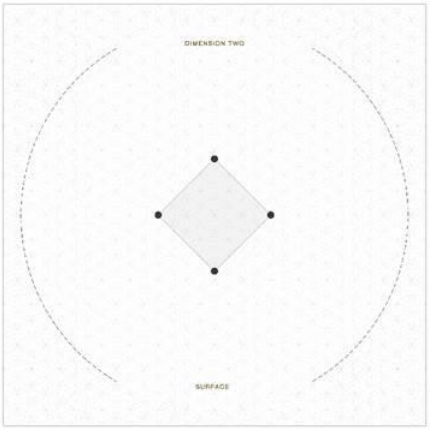
As mentioned earlier, Bragdon, in his theories on four-dimensional perspectives, drew upon the ideas of Charles Howard Hinton, a mathematician, and writer, in the visualization of higher-dimensional geometry.

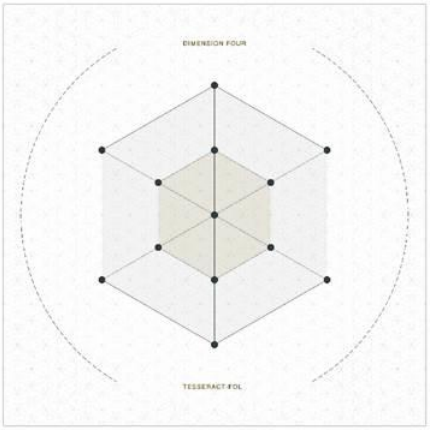
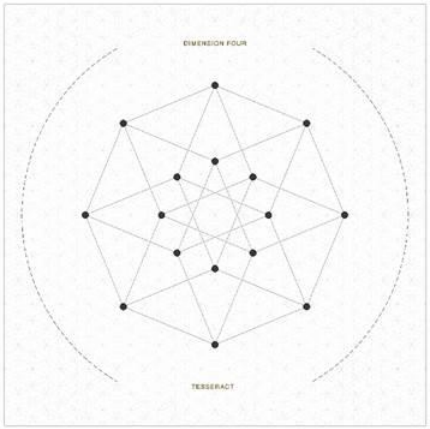
Studying space without emphasizing time was the prevailing notion in the period before Einstein. However, for a more comprehensive understanding of the unified awareness, one must cohesively examine the concept of space-time. (Critchlow, 1969, p. 3)

The fundamental element of this universe is space, which is inherently empty and void. For this reason, it can encompass everything within itself. In theoretical terms, space (empty space) is considered the primordial condition for the existence of anything. (Govinda, 1961)

With this background, we can discuss abstract patterns of sacred geometry, how they evolved, and the stories and philosophies transmitted at each stage. While the presence of this geometry may not be overt in today's world, a return to nature reveals its tangible and intangible presence everywhere.

Table no.1 - Foundations of Spacetime Geometry (Geometric Forms in Mathematical Dimensions), Drawings by the Author

Figure Name	comments	Figures
DOT	- zero dimension - only can specify a point in the coordination systems - (x, y, …) for two dimensions and further.	
LINE	- one dimension - created from connecting two points - straight line; y=mx+b | single degree equation - its only parameter is the length	
PLANE	- two dimensions - created from three or more lines - planar plane; A(x−a) + B(y−b) + C(z−c) = 0 - two parameters to discuss: perimeter & and area	

CUBE [volume]	- three dimensions created from 6 (at least 4 for pyramid) planes three parameters to discuss: area & height & volume - $\boldsymbol{V} = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$	
HyperCube [tesseract]	- four dimensions - created from six cubes - parameter to discuss: time - $\boldsymbol{V} = \begin{vmatrix} A_x & A_y & A_z & A_i \\ B_x & B_y & B_z & B_i \\ C_x & C_y & C_z & C_i \\ D_x & D_y & D_z & D_i \end{vmatrix}$	
Alternative Projection Of HyperCube [tesseract]	- another projection of the tesseract	

Introduction to Sacred Geometry:

"In the beginning, there was 'Silence and Stillness,' and there was nothing. Silence desired and moved. We call it 'Aether' or Spirit. The Spirit, desiring, set things into motion, and through its movement, the first essence in existence was created, which was space. After the movement, existence needed to emerge from the darkness. For this reason, in the second stage, the creation of light was added to make the essence visible. In the third stage, duality was overcome, and balance was established. The desire continued to bring forth the initial seed of life." (The First Structure in Nature)

Our ancestors held a deep reverence for the 'Seed of Life', also known as Genesis Pattern, as it was believed to be the source of all existence. This sacred symbol represented the very origin of the world we know as 'Reality', and was held in high regard for its profound meaning and significance.

The distance between the centers of the circles (assuming the unit radius as "1") is equal to the square root of three. If we draw lines passing through the centers of the first and second circles to one of the intersection points, its proportions with the Pythagorean theorem formula indicate that the triangle is right-angled. The next circle is then drawn at the intersection of the two previous circles, and this path leads us to the geometry of the seed of life.

If we continue with the seed of life, this pattern becomes more complete, leading to the next geometry we encounter, known as the "Flower of Life" (a network of overlapping circles). This can be considered the starting point of human geometry. This geometry and its surrounding

shell can be seen as the first representation of the human perception of the cellular membrane. (Lundy, 2010, 70; Oscar, 2018, 122)

The geometric form we refer to as the "Seed of Life" is, in fact, a reference to the first set of awareness cells that form the embryo. Essentially, it represents the initial set of genetic information that shapes and evolves into what biologically defines us today through this pattern.

Now, if we further elaborate on the evolutionary process of this pattern (by removing the Flower of Life shell[4] and replicating the pattern through the continuation of the previous method, i.e., by adding circles to the intersection points of the previous circles), we arrive at a more complete and comprehensive geometry. This geometry seems to mimic the living nature, growing and, with each stage of completion, introducing more elements and creating more intricate form beauties. (Lundy, 2010, 72)

The next geometric form in this evolutionary path is called the "Fruit of Life," and it can be regarded as the fabric of reality surrounding us. This fundamental geometry, from which Plato derived the regular polyhedra as the elementary elements of three-dimensional geometry, is extracted. Pythagoras, in his geometric studies, considered each of these five polyhedra as a symbol of the elements shaping existence. However, certain findings in artworks, such as carved stone balls from the Neolithic era in Scotland, suggest that these objects were known for a long time, it is challenging to accept that the purpose of creating these artifacts at that time was for the widespread understanding of mathematics. Instead, they were likely crafted for decorative or religious reasons.

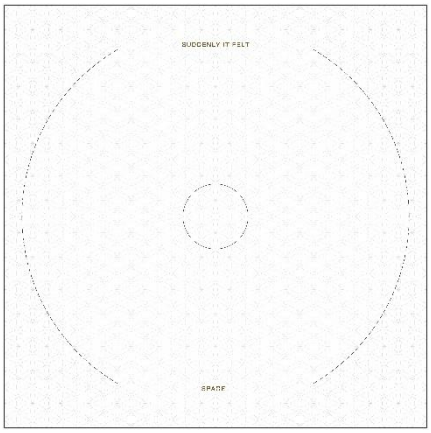
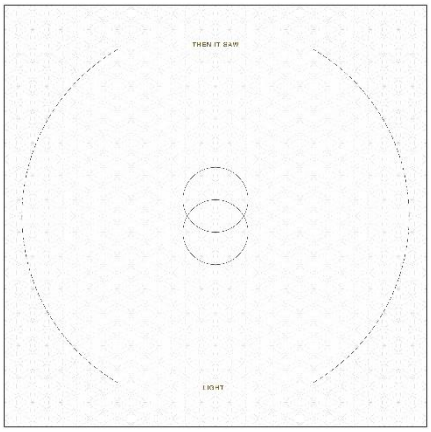
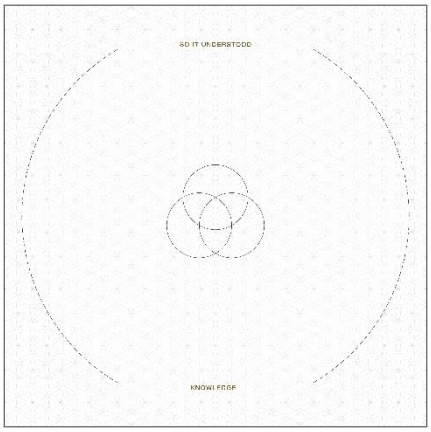
4 - Zona pellucida

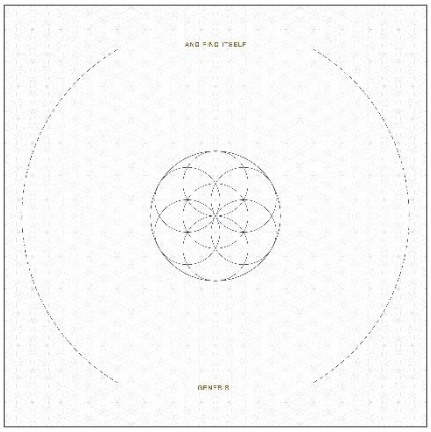
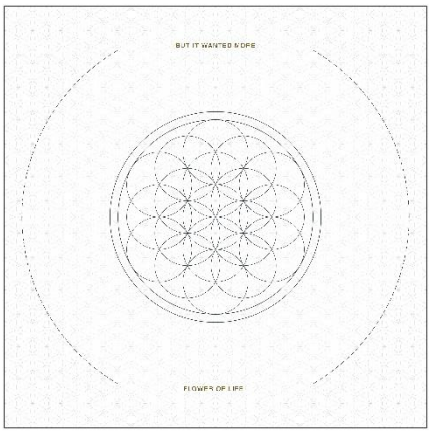
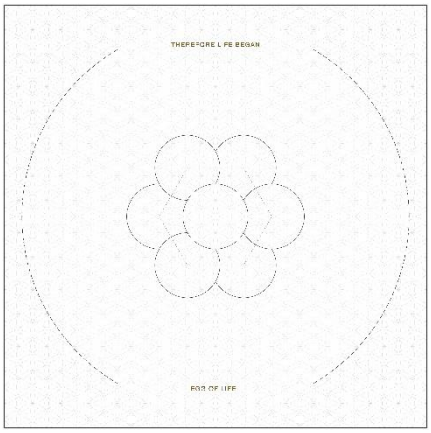
The initial idea behind these artifacts was drawn from Euclidean principles, but interpretations of them have consistently gone beyond the realm of pure mathematics into the domains of philosophy and theology. This broader understanding can be traced back to the time of Plato.

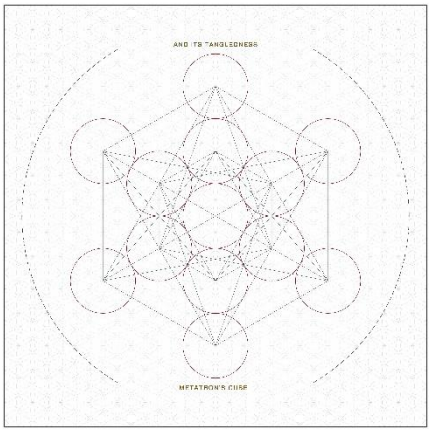
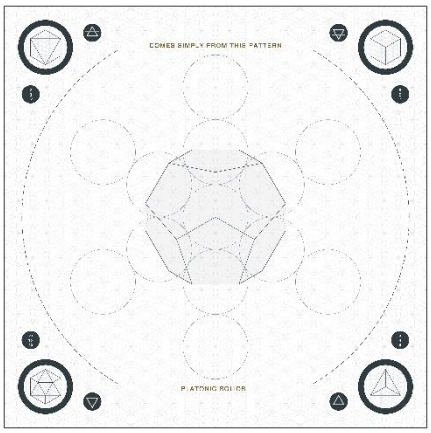
Plato, in his work "Republic," expresses that *"geometry draws the soul towards truth."* Later, in the "Timaeus," he mentions Plato's solids for the first time, describing them as the most beautiful structures possible. In this dialogue, Plato establishes a strong connection between these solids and foundational philosophy, emphasizing the influential role of sacred geometry in the emergence of these forms.

In the second table, we observe the method of enclosing and drawing these five Platonic solids in the pattern of the "Flower of Life."

Table no.2 – Sacred Geometry Patterns, Drawings by the Author

Figure Name	comments	Figures
First Circle	- space - the very first object - she moved and with her movement, she created the space to find herself.	
Vesica Piscis	- light - the second state was creating the light to see the object and give an end to darkness	
Tree Circles [trinity]	- Triquetra - the balance - they also call it the knowledge	

Six Circles [seed of life]	- the first element of the creation - the first geometry	
Flower of Life	- the pattern that shows us the very first elements of life and reveals the secret of the Tree of Life	
Egg of Life	- the first three-dimensional geometry which we can find a cube hidden inside its center - the original 8 cells in biology	

Fruit of Life [Platonic Solids]	- the mother geometry which also represents Metatron's cube - the basic patterns that we can extract platonic solids from it	
Platonic Solids extracted from Fruit of Life	- the very first elements in alchemy - tetrahedron is fire - hexahedron is earth - octahedron is air - icosahedron is water - dodecahedron is ether	

Platonic Solids

Plato (428/427 BC) refers to the five regular polyhedra as the most beautiful structures and attributes them to the four primary elements in alchemy (fire + earth + air + water). However, he describes the fifth element as follows: *"The fifth element continues to exist while not made from any substance or matter, and God has used it for the division of the zodiac (celestial spheres)."* Later, Aristotle, a student of Plato, names the fifth element "aether." (Hejazi, 2008, 7)

Platonic solids are examined in the thirteenth book of Euclid's Elements from the perspective of pure mathematics, focusing on structures and proportions. This serves as a fundamental reference for many mathematical studies in the second millennium.

In Timaeus, Plato describes creation as being based on five fundamental shapes, known as "Platonic solids." These are the only possible regular polyhedra whose faces are congruent regular polygons. (Skinner, 2007, p. 55) (Lundy, 2010, p. 132)

Many philosophers of that time discussed sacred geometry, and one of the most well-known examples of metaphysical beliefs regarding this geometry is attributed to Pythagoras. Their ideas about sacred geometry, particularly those of Plato in the 5th century BC, have been highly influential. In "Timaeus," Plato writes: *"Geometry rightly treated, is the knowledge of the eternal" and "draws the soul towards truth and creates the spirit of philosophy."*

In "Timaeus," Plato establishes a strong connection between geometry and the essence of philosophy. He discusses five regular polyhedra, which have regular faces and congruent sides, naming them "tetrahedron," "octahedron," "cube," "icosahedron," and "dodecahedron." (Tavakoli, 2020, p. 2)

Other perspectives from ancient Egypt and beyond also exist, associating the fifth element with the void and Antimatter. Aristotle, his disciple, added the fifth element and called it "aether." Historically, due to the golden ratio, aether is often linked with the dodecahedron, which, from a purely mathematical perspective, is the focus of the thirteenth book of Euclid's Elements, where the study of proportions and their structures is explored within a circle. (Tavakoli, 2020, p. 2)

In Timaeus, Plato considered earth, air, fire, and water as the fundamental elements of the world. He equated Platonic solids with these four elements: the cube with earth, the icosahedron with water, the tetrahedron with fire, and the octahedron with air. (Hejazi, 2008, p. 7)

Plato writes:

"Now we want to distribute the four shapes that we described among the four elements, namely fire, earth, water, and air. Earth has the shape of a cube because, among the four elements, it is the most immobile. A body with such properties must necessarily be based on more stable principles. We assign the heaviest one (icosahedron) to water, the lightest one (tetrahedron) to fire, and the average one (octahedron) to air."

Plato writes about a specific fifth shape:

"Still, one combination remains, which God used to adorn the celestial forms in the sky."

By adorning the celestial figures, he meant the connection of the twelve pentagonal faces to the twelve zodiac signs, encompassing the entire cosmos. (Hejazi, 2008, p. 7)

The Platonic solids have garnered significant interest for various reasons, including their practical applications in modern mathematics, their presence in the crystalline structure of minerals, and their role in scientific models explaining metaphysical phenomena. Today,

mathematicians and contemporary scientists appreciate these solids for their elegance and the profound insights they offer. Moreover, visual arts, have captivated audiences due to their inherent beauty and philosophical significance.

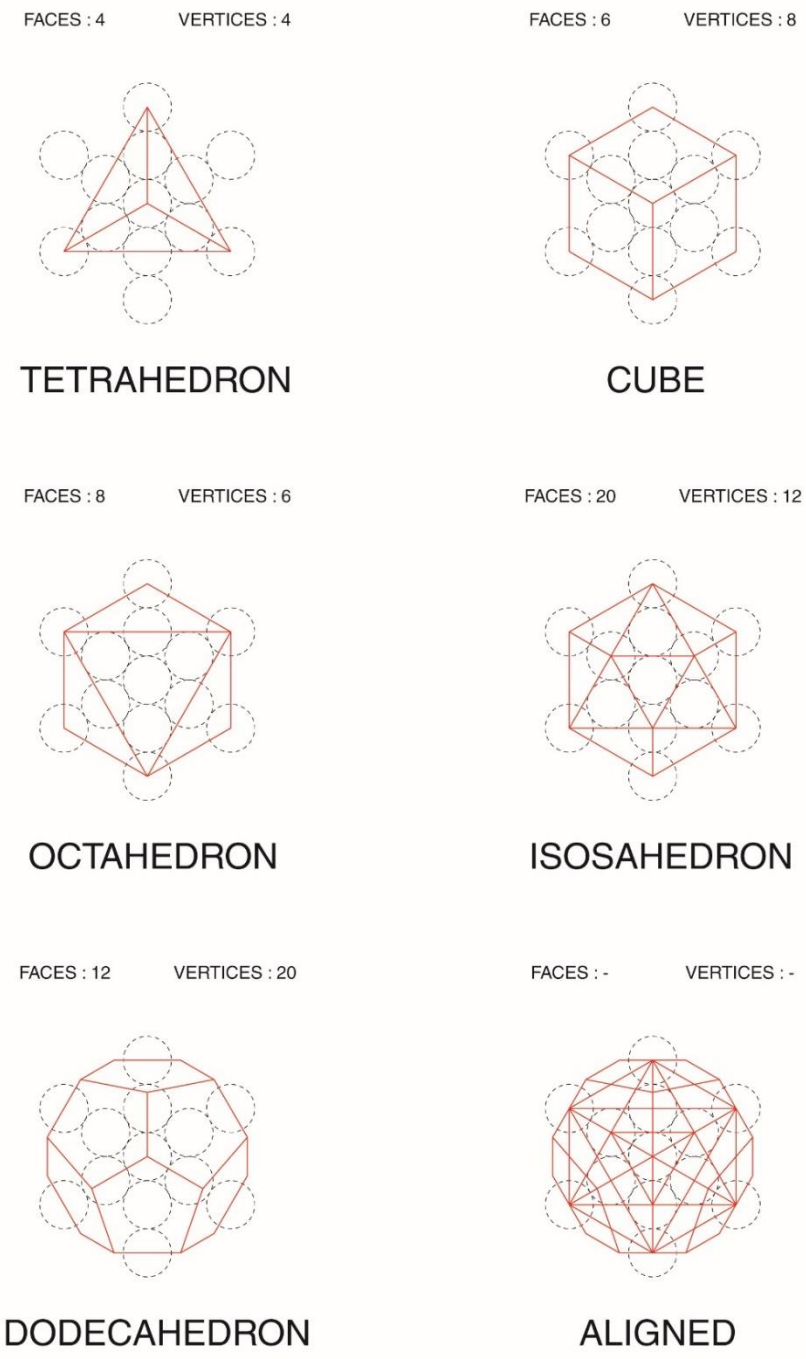


Figure No.1 – Drawing platonic solids into the Fruit of Life pattern (Hejazi, 2008), Edited by the Author

Golden Ratio

The Golden Ratio, denoted by the symbol Φ, is a mathematical concept expressed by the ratio $1 = \frac{1+\sqrt{5}}{2}$ or approximately 1.618, It is evident in the proportions of various natural phenomena such as plants, flowers, viruses, DNA molecules, seashells, planets, and galaxies. The ancient Athenian sculptor Phidias (circa 480 – 430 BC) is believed to have incorporated the Golden Ratio into his works of art. This ratio is unique in that when the larger part is divided by the smaller part, it is equal to the ratio of the sum of the two parts to the larger part. The Golden Ratio symbolizes rebirth, ascension, and expansion from unity, as each birth is linked to the existence preceding it. It represents the manifestation of the complete division of unity. (Hejazi, 2008, p. 8)

The Golden Ratio is considered the most aesthetically pleasing proportion in beauty aesthetics. Throughout the history of art and architecture, traditional artists have embraced the divine ratio as a sacred measure and a principle of aesthetic proportion to embody the spirit in material form. This ratio is also observed in the design of the Great Pyramid of Egypt. (Hejazi, 2008, p. 8)

Muslim scholars initiated their studies in geometry by exploring Greek works, particularly those of Euclid and Apollonius. This engagement began in the early Abbasid Caliphate period when they became acquainted with these works. It is noteworthy that a significant focus on the science of geometry in Baghdad was spurred by the encouragement of the Banu Musa brothers, especially through the composition of their book "Ma'rifat Masah al-Ashkal" (Knowledge of the Measurement of Figures). Nasir al-Din al-Tusi later provided commentary on this work. The book was translated into Latin and had a substantial impact on Fibonacci and Thomas Bradwardine. (Nasr, 1978)

"Sacred geometry in Islamic art also incorporates numbers and ratios that facilitate a movement towards unity through various divisions and shapes." (Kamrani, 2000, p. 114)

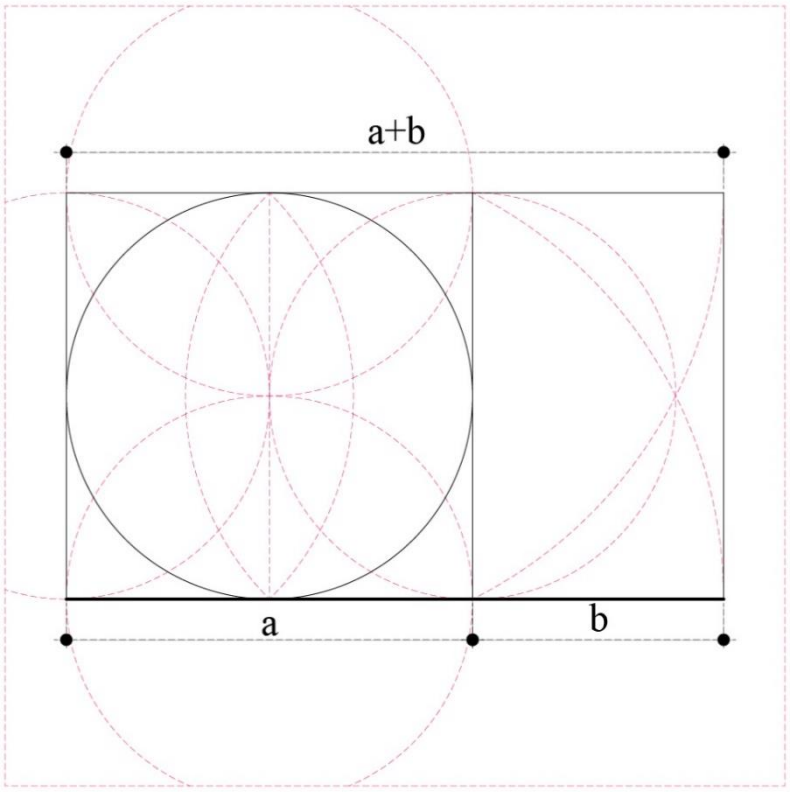


Figure No.2 – Technical drawing of Golden Ratio (Hejazi, 2008), Edited by the Author

Sacred Art

"I saw the celestial climates opened up to the highest heavens; I saw the cherubs and spirits in such a way that it seemed their descent to the world brought forth a sigh of regret." By Ruzbehan Baqli (Kamrani, 2000, p. 145)

The history of religious art can be seen as a reflection of humanity's imagination, aspirations, and understanding in the face of the divine power.

"The styles have remained clearer and more transparent the closer they have been to the source." (Kamrani, 2000, p. 147) This criterion also holds for artists; some artists who seem to play the role of intermediaries between artistic currents, the closer they are to the essence, the more profound and the more mysterious their works become.

The unattainability and the silence prevailing over the forms and endless mysteries present in it have made sacred art more indicative and closer to the truth. For this very characteristic, sacred art is usually practiced by artists who themselves are mystics and seekers. (Kamrani, 2000, p. 150)

With the help of the history and anthropology of religions, one can access some of the symbols hidden in sacred art. The iconic depictions in early Christianity, calligraphy and patterns in Islamic art, and paintings from various schools in the East (especially the Daoist school) are prominent examples of sacred art.

Sacred Art in Iran

Before the advent of Zoroastrianism, Persians used to sanctify the angels and deities of good, who were commanded by the one true God. (Kamrani, 2009; Hajari, 2011, p. 19)

Zoroaster strived to return the Persian tribes from multiple gods to a single, unified God. In the Zoroastrian religion, Ahura Mazda is the light of all lights, meaning all the lights of the world emanate from his divine status, and his divine commands are executed by the eternal assistance of six holy angels, known as Amesha Spentas[5]. (Hajari, 2011, 19)

The attachment to the Farahi religion has been a characteristic of the Persian spirit, which after Islam, manifested and reached perfection in the sacred meaning of the Sufi path of guardianship. Sacred geometry in ancient Iran, before the advent of Islam, was influenced by the Mazda religion.

In the Mazdaean tradition, similar to other Eastern dualistic religions, there exists a dualistic belief system rooted in a metaphysical reality. Cosmology in this tradition is divided into two concepts: Mihr (*Mithra*[6]) and Geeti (*Geithi*[7]), which are introduced in Pahlavi[8] texts as attributes and nouns. Here, the noun signifies cosmological concepts, such as spiritual and invisible entities, and material and visible entities, while the attribute refers to the classifications of spiritual and material beings (Bois et al., 2008, p. 69; Asadi, 2016, p. 23). Thus, the structure of beings is generally expressed in two senses: first, in the dual aspects of material and spiritual, and second, in the hierarchy of beings.

5 - In Zoroastrianism, the Amesha Spenta is a class of seven divine entities emanating from Ahura Mazda, the highest divinity of the religion. (“Amesha Spenta”, 2023)
6 - Indo-Persian divinity
7 - Universe
8 - Pahlavi is a particular, exclusively written form of various Middle Persian languages. (“Pahlavi scripts”, 2023)

The manifestation of matter in the Mazda religion means that Ahura-Mazda[9], along with the physical body, has also created the Fravashi, the spiritual part of each human being (the luminous body). (Nasr, 2006, 55) Each earthly creature has a celestial or *Minak*[10] counterpart, and whatever the Geithi (earthly aspect) does is the development of the reality of Minak through the evolution of the soul and imagination. This evolution (here meaning the action and change of the soul) is not possible without the aid of the angelic light, which the Zoroastrian religion refers to as *Khorneh* or the light of glory. (Asadi, 2016, 23)

This light is initially considered a sacred light that brings order and harmony to beings. In art, this light is depicted as a luminous halo. In Bundahishn[11], this light is considered one with the essence of the soul. Khorneh is a soul that makes our connection from the earth as a purely material object to the world of angels as a metaphysical world possible. (Nasr, 2006, 129) In fact, Khorneh with its angelic glory signifies the unity and connection of earth and heaven in the Zoroastrian religion. Khorneh is the same light, glory, and angelic glory that exists at the center and within everyone and is a manifestation of the soul. This center is the focus of creativity and connection to the center of existence and the inner world. Khorneh and the six countries around it (the image of the seed of life) are themselves a kind of mandala and a world map. We can also find Khorneh in Islamic mysticism, the eighth climate, and Jabulqa and Jabulsa[12] and Hurqalya[13], refer to this mythical space which is the same as the world of example. (Razi, 1992, 129)

9 - Also known as Asura Mazda, Hormaz, and Hurmuz, is the creator deity and the god of the sky in Zoroastrianism.
10 - Celestial Aspect (“Ahura Mazda”, 2023)
11 - Bundahishn is the name traditionally given to an encyclopedic collection of Zoroastrian cosmogony and cosmology written in Book Pahlavi.
12 - Jabulqa and Jabulsa (Arabic: جابلقا وجابلسا) or Jabalq and Jabars (Arabic: جابلق وجابرس), are two legendary cities mentioned in Islam. They are said to be made of emerald and visited by the Islamic prophet Muhammad in his Night Journey. (“Jabulqa and Jabulsa”, 2023)
13 - The realm of Malakut (Arabic: عَالَم الْمَلَكُوت, 'world of the kingdom [of God]'), also known as Hurqalya or Huralya, is a proposed invisible realm of medieval Islamic cosmology. (“Hurqalya”, 2023)

The connection between sacred geometry and sacred art in Eastern and Western philosophy

Historians and philosophers disagree about the meaning of this form of knowledge that Suhrawardi[14] derived from the combination of the wisdom of Plato and Zoroaster and named it Illumination. While Abdolrazagh-Kashani[15] considers him a follower of Seth[16], who had a close connection with the Hermetism[17] school, Ibn-Wahshiyya[18], who was the first person in the Islamic world to use the term Illumination, considers the Illuminators a class of Egyptian priests who were the children of Hermes' sister. (Nasr, 2009)

The term "Ishraq" in Arabic is related to the word "Sharq" (East) and conveys the meaning of light and luminosity. Similarly, Suhrawardi, based on these meanings, utilizes the term "Ishraq" along with its symbols and allegories to describe his world. In this philosophy, "Mashriq" (East) signifies the realm of pure light or the world of angelic entities, while "Maghrib" (West) represents darkness or matter. In between these two realms lies the intermediary realm "Maghrib-Wasat" (Central West), which corresponds to the celestial spheres and the heavens, where light and darkness intertwine, giving rise to the visible world.

In Suhrawardi's time, two distinct philosophical perspectives were evolving in the East, significantly different from other philosophical views. The philosophy of Ishraq is one of those perspectives, established by Suhrawardi based on a form of intuitive thinking. According to Suhrawardi, Aristotle does not hold philosophical centrality because philosophy is the pursuit of

14 - "Shihāb ad-Dīn" Yahya ibn Habash Suhrawardī was a philosopher and founder of the Persian school of Illuminationism.
15 - Islamic Philosopher in the 7th Century in The Solar Hijri (~1320 AD)
16 - Ancient Egyptian God of Chaos & Storms
17 - Hermeticism or Hermetism is a philosophical and religious system based on the purported teachings of Hermes Trismegistus (a Hellenistic conflation of the Greek god Hermes and the Egyptian god Thoth) ("Hermeticism", 2023)
18 - Ibn Waḥshiyya, died c. 930, was a Nabataean agriculturalist, toxicologist, and alchemist born in Qussīn, near Kufa in Iraq. ("Ibn Wahshiyya",2023)

truth, and the truth is not confined to the teachings of Aristotle. The second perspective was *Islamic mysticism and gnosis*[19], which reached its pinnacle with Ibn-Arabi[20]. According to the Ishraq school, attaining the truth requires not only logical reasoning and intellectual arguments but also the experience of asceticism, self-purification, and inner purity. The philosophers of Ishraq were followers of Platonic theories. (Hajari, 2011, 19)

Plato pays considerable attention to the geometry of Pythagoras and the music of the spheres. In "Timaeus," he elucidates the geometric creation of the world, stating that the Creator has fashioned the visible world in resemblance to a geometric ascent. In "The Republic," he writes about geometry, saying, *"Geometry; its recognition is the apprehension of that which never becomes different, not the apprehension of entities that are the subject of time, appearing now and vanishing at another time. Geometry propels the soul toward the truth and nurtures a wise spirit within humans"* (Republic, VII).

Plato's devotion to geometry is so profound that it is reported to have been inscribed above the entrance to his academy: *"Let no one ignorant of geometry enter here"* (Iyuz, 2004, p. 106).

Suhrawardi and other followers of the Illuminationist philosophy were not confined to mere rational arguments and proofs. They were people of unveiling and intuition, delving into the secrets of the heart. Illuminationist wisdom is a conscientious observation and a mystical contemplation, through which the truth of existence manifests itself to humanity as it truly is. Suhrawardi's horizon of perception and intuitive panorama was immensely broad. From his

19 - Sufism may be best described as Islamic mysticism or asceticism, which through belief and practice helps Muslims attain nearness to Allah by way of direct personal experience of God.
20 - Ibn ʿArabī was an Andalusi Muslim scholar, mystic, poet, and philosopher.

perspective, the predecessors surpassed their contemporaries in their connection with reality and their understanding of the truth, illuminated in a manner more profound and lucid.

Thus, In the philosophy of Illuminationism, which is a combination of Platonic and Zoroastrian ideas, the world is called the world of example or the celestial earth of Hurqalya, and it is the same world that is called the true existence, and the world of appearances and examples is placed in the position of angelology, and its subject is to reach the emerald stone and step into the world of angels. (Corbin, 2004, 21-22)

According to these principles, one can perceive the mandala as the very point and inner essence of an individual. The circles within it represent the worlds and realms surrounding it. This luminous particle bestows a sacred order and structure upon beings, and the center within each person signifies the sacred geometry of the individual.

After several decades, the teachings of Suhrawardi became widespread among philosophers in Iran, the Middle East, and India. When the Safavids came to power in Iran, the influence of Illuminationist teachings reached its peak.

During the Safavid era, Shia Islam became the official sect in Iran, and not only did Persian art and architecture flourish, but also rational sciences experienced a renaissance. Unlike art, which is widely recognized in the world, this intellectual revival in Safavid Iran remained largely unknown outside the country until contemporary times. (Hajari, 2011, 51)

Mir-Emad[21], the greatest teacher of his time during the reign of *Shah-Abbas*[22], revived the discussion about the philosophy and wisdom of Illuminationism. After Mir-Damad, his most

21 - Mir Emad is perhaps the most celebrated Persian calligrapher and philosopher.
22 - Abbas I, commonly known as Abbas the Great, was the fifth shah of Safavid Iran from 1588 to 1629.

famous student, *Mulla-Sadra*[23], who is considered one of the greatest Muslim philosophers of all time in Iran, recompiled Suhrawardi's theories and perfected them. Mulla-Sadra's beliefs included elements of *Mashai*[24] philosophy in addition to Suhrawardi's theories, and some of the principles of Ibn Arabi's mysticism were also included in it. Suhrawardi's works gradually came under the spotlight of official schools of that time under the influence of Mulla-Sadra's writings, and Illuminist beliefs were studied and researched again. Islamic mysticism and Sufism, along with Suhrawardi's Illuminationist philosophy, were completed by Ibn Arabi. (Hajari, 2011, 51-52)

Realm of imagination

The Imaginal World, or the World of Archetypes, introduced by Suhrawardi, is a highly intermediary and liminal realm. According to Suhrawardi's interpretation, the imaginary world lies between the two realms of the rational and the sensible. It is neither entirely corporeal nor purely spiritual but incorporates characteristics from both realms within itself. (Suhrawardi, 1879, 64-65)

Henry Corbin emphasizes the creative aspect of the Imaginal World, suggesting that imagination has the power to transform the world of origin. In this sense, imagination can perceive realities in sensible beings that other mental faculties are incapable of. For instance, it can bring rational beings (the material world) into the realm of imagination and envelop them in a unity perceivable by the sensible mind. This ability solely belongs to the realm of imagination. (Corbin, 2004; Hajari, 2011, 63)

23 - Ṣadr ad-Dīn Muḥammad Shīrāzī, more commonly known as Mullā Ṣadrā, was a Persian Twelver Shi'i Islamic mystic, philosopher, theologian, and ʻĀlim who led the Persian cultural renaissance in the 17th century.
24 - Peripatetic school;

The Imaginal World can be considered the very source of creation. It is the moment when the soul decides to move, and the geometry of creation takes shape.

Corbin also alludes to the idea that God created the world through the faculty of imagination. The Imaginal World is the realm where spirits take form and bodies find abstraction. Here, the Imaginal World reigns supreme. (Corbin, 2004; Hajari, 2011, 65)

In the mysticism of Ibn Arabi, the initial image is such that creation is a kind of manifestation. This implies that the act of creation is the actualization of the divine imaginative power. (Hajari, 2011, 65)

Islamic Geometric Patterns and Arabesque motifs

Rosette patterns, constituting the main elements of dome designs, are known today by various names such as Islamic, "*Khatayi*[25]", and interlacing knot patterns. These patterns can be observed in artifacts across different regions and cultures over time.

Islamic and Khatayi patterns, which constitute a significant part of Persian visual art, were initially known by different names. The exact terms used to refer to them before Islam are not entirely clear, but the term "swastika[26]" was somehow associated with the movement of Islamic patterns. (Lotfi, 2010, p. 80-81)

A specific term attributed to these patterns is not precisely defined, but from the 16th century AD, when Europeans became interested in Islamic art, the term "Arabesque" was used to refer to such patterns. (Pakbaz, 2000, p. 79)

One of the main features of these patterns is the principle of continuity. According to this principle, these patterns are seen continuously and consistently over time or even throughout a single artwork.

Repetition is another principle crucial in the art of Persian imagery, serving as a key tool in the creation of many patterns. Specific forms are repeated in patterns, and significant changes are rarely observed over centuries. It seems that these patterns provide continuity to certain arrays and ancient traditions. Some of these forms are sometimes presented as conventional rituals and are seldom subject to change. (Lotfi, 2010, p. 88)

25 - Type of Islamic arabesque based on organic forms. (Floral patterns)
26 - The swastika is an ancient religious and cultural symbol, predominantly in various Eurasian, as well as some African and American cultures.

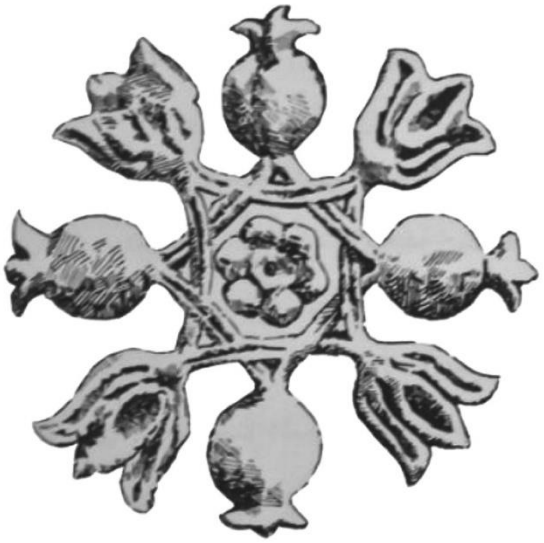


Figure No.3 – Example of a repeating pattern from a prominent stucco design of Tisaphernes, Sasanian period. (Lotfi, 2010), enhanced by the Author

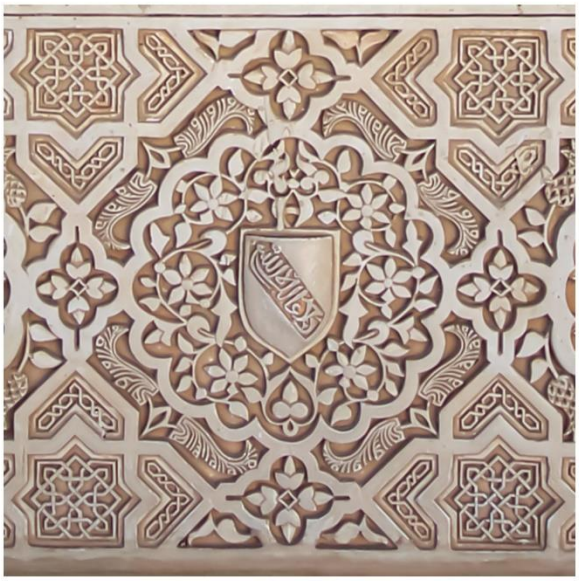


Figure No.4 – A part of the decorations (stucco) of Alhambra in Granada, Metropolitan Museum of Art collection, enhanced by the author.

Spiral motifs

As mentioned earlier, both Islamic and Khatayi patterns are primarily composed of helical and spiral structures. If Islamic branches are added to this structure, it forms an Islamic pattern, and if flower and leaf patterns of Khatayi are added, it is called the Khatayi movement.

The traces of spiral patterns can be observed in goat horns in ceramic artifacts from the Neolithic era (approximately nine thousand years ago). After the invention of the potter's wheel, significant developments occurred in these patterns, and ceramics were adorned not only with simple geometric patterns but also with more elaborate depictions of plants, animals, and humans. (Reyazi, 1996, p. 89)

The Fibonacci sequence gives rise to one of the most famous spirals, highly revered by Western artists, especially during the Renaissance. This pattern has been widely used in architectural proportions and compositions, both in Christian and Islamic art, symbolizing cosmic vitality and birth. In more evolved forms of these patterns, intricate and distinct lines are observed,

forming the fundamental structure of Islamic geometric patterns. It seems to start from a geometric center and intricately move outward. As mentioned earlier, the meaningful connection of these patterns with natural spirals constantly reflects the themes of recurrence and growth found in nature. Additionally, one can observe movement and coherence reminiscent of natural forms within these patterns.

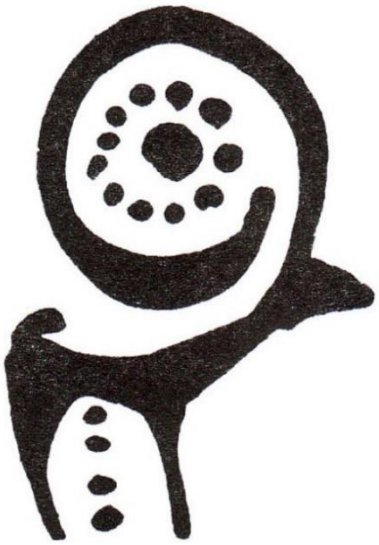


Figure No.5 – A sample of a spiral pattern in painting on pottery, 4th millennium BC (Lotfi, 2010), enhanced by the Author



Figure No.6 – A sample of a spiral pattern in the painted design on incised buff pottery, Susa civilization. (Lotfi, 2010), enhanced by the Author

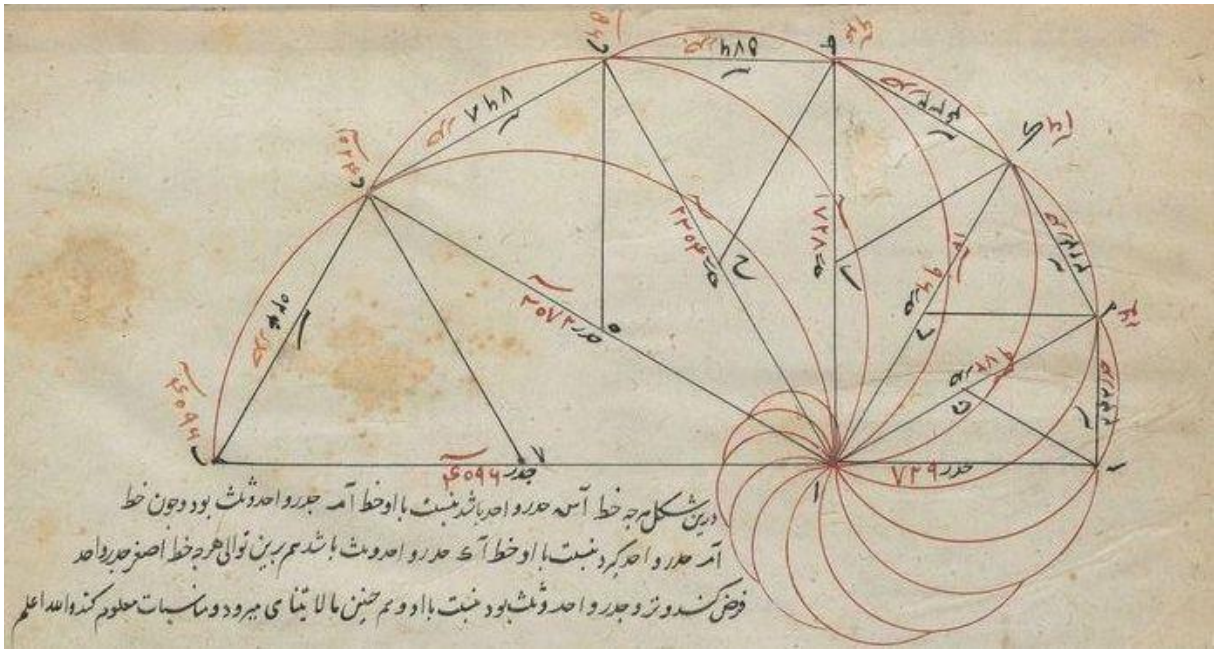


Figure No.7 – Technical drawing of the spiral form (Fibonacci sequence) in the handwritten texts of Al-Karaji in the book "Anbat Al-Miyah Al-Khafiya."[Extraction of hidden waters] (Hatami, 2021), enhanced by the Author

Islamic arabesque and Floral motifs

Initially, these patterns were predominantly used in calligraphy and the embellishment of manuscripts, such as silk patterns and geometric backgrounds in decorating the pages of the Quran and other scientific and philosophical books. During the Safavid period, with the flourishing of Islamic art in Iran, these patterns became widespread in tilework and architectural art. The architecture of mosques, palaces, and mausoleums all embraced these rich patterns, and even urban geometry drew inspiration from the constituent patterns of these designs.

An example of Islamic patterns can be observed in the movement of spiral patterns from the outside to the inside. Similar to the floral patterns, these patterns follow the same scheme found in the natural patterns of plants, however, with the distinction that they diverge significantly from their initial form and evolve through various mental processes. The commonality between Islamic and floral patterns lies in their adherence to a unified pattern.

The use of botanical and animal patterns, as mentioned, dates back to ancient historical periods. However, the incorporation of botanical patterns derived from the *Tree of Life*[27] holds a special significance. As far as we know, the palm tree, a type of the Tree of Life, and its leaves have played a pivotal role in shaping Islamic and knot patterns. (Lotfi, 2010, p. 99-100)

The Tree of Life is a common motif in Western Asia, appearing in various forms on cultural artifacts and products of that time. It is often depicted as two goats, two humans, or two mythical beings standing on either side of the tree. This motif symbolizes the celestial tree and heavenly values, the symbol of life and renewal, immortality, and the center of existence. This symbol can

27 - The tree of life is a fundamental archetype in many of the world's mythological, religious, and philosophical traditions. It is closely related to the concept of the sacred tree. (“Tree of life”, 2023)

be identified with the *Haoma*[28] tree in Zoroastrianism. The representation of this motif can be traced from the Sumerians to Sassanian art and onward to the art of the Islamic period. (Reyazi, 1996, p. 66-67)

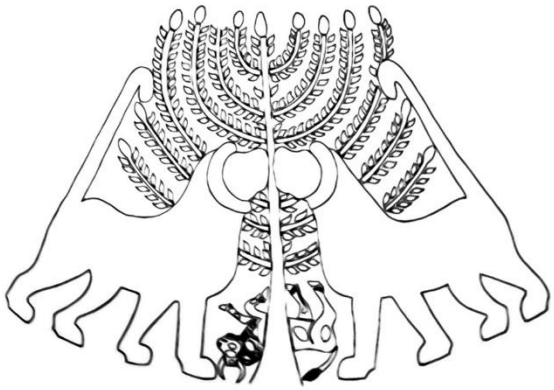


Figure No.8 – Tree of Life pattern on found clay vessels in Lorestan. (reconstructed pattern), (Lotfi, 2010), enhanced by the Author

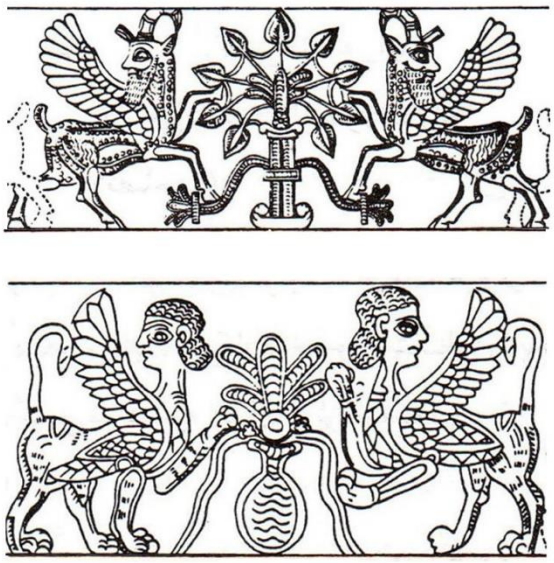


Figure No.9 – Tree of Life pattern on stone vessels in the civilization of Jiroft civilization. (reconstructed pattern), (Lotfi, 2010), enhanced by the Author

The use of the fruit and leaves of this tree held significant value in the lives of people of that time. Images of this tree, its leaves, or its fruit can be observed in the civilizations of Southwest Asia from prehistoric times to the art of Iran. (Reyazi, 1996, p. 208)

In samples from the *Sasanian*[29] period, a pattern resembling half a palm leaf can be observed. This pattern has undergone a gradual transformation influenced by various factors, including the removal of some leaf teeth, changing straight lines and sharp-pointed forms to curved lines, ultimately evolving into a pattern that can be considered one of the motifs of Islamic designs. This evolutionary trend continues from the Sasanian period to the Islamic period, where these

28 - Haoma is a divine plant in Zoroastrianism and in later Persian culture and mythology. Haoma has its origins in the Indo-Persian religion and is the cognate of Vedic soma.

29 - The Sasanian Empire or Sassanid Empire, officially known as Eranshahr also known as the Second Persian Empire, or Neo-Persian Empire was the last Persian empire before the early Muslim conquests of the 7th–8th centuries AD

patterns take on imaginary forms and deviate significantly from their source. The images demonstrate this evolutionary process.



Figure No.10 - The use of palm leaves in a section of architectural decorations (stucco) in Tisfoon, Sasanian period., Metropolitan Museum of Art collection, enhanced by the author.

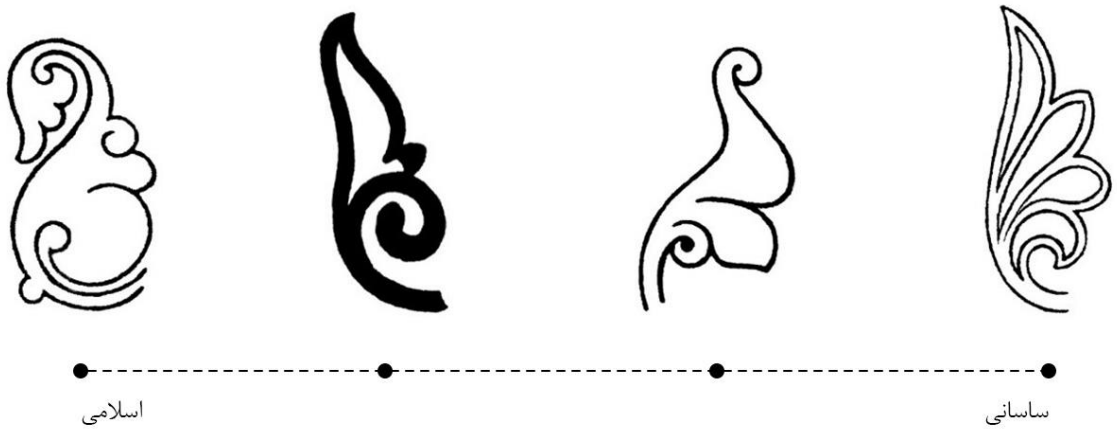


Figure No.11 – A visual example of the evolutionary transformation of the palm half-leaf from the Sasanian period to the Islamic period, (Lotfi, 2010), enhanced by the Author



Figure No.12 – Islamic and Khatayi patterns on the outer surface of the dome of Sheikh Lotfollah Mosque, Safavid period, Photo & edit by the author.



Figure No.13 – A portion of Islamic and Khatayi patterns on the outer surface of the dome of Jameh Abbasi Mosque (Imam Mosque), Safavid period, Photo & edit by the author.

With the advent of Islam and the relative restrictions on depicting human figures, artists shifted their focus more toward floral and vegetal patterns. Drawing patterns of flowers and plants became a conventional method for decoration during this period. (Wilson, 1998, p. 143; Lotfi, 2010, p. 140;)

Shah Abbasi floral patterns (Lotus, Nelumbo nucifera)

Nelumbo nucifera, also known as lotus, which gained fame as the Shah Abbasi flower during the Safavid period, has a long history in the culture and civilization of Iran and the world. After the advent of Islam, much of this pattern received attention and imitation from Persian artists with the introduction of Chinese ceramics and artistic techniques. (Wilson, 1998, p. 138) However, even before Islam, this pattern can be observed abundantly in various forms among the works of Persian artists. The design of the lotus flower, which has attracted the attention of Persian artists after Islam, is generally an imitation of Indian and Chinese patterns. Still, before Islam, it was more often inspired by Egyptian designs.

Bearing the lotus from the water symbolizes blooming from the chaotic world and its geometric beauty symbolizes the order and magnificent beauty of the celestial world. The opening and closing of the lotus also represent the manifestation of light from the Creator of existence. (Bolkhari, 2005, p. 8) In this way, the role of the lotus flower is a unique symbol of the mysteries of nature in all ancient religions.

In Persian culture, the lotus flower holds a special place, and ancient Persians even organized a celebration by this name. In Persian mythology, the symbol of the water goddess Anahita is the lotus flower. (Lotfi, 2010, p. 129)

The symbolism of the lotus flower can be traced to Egyptian myths, where the famous myth of *Hermopolis*[30], which revolves around water, the sun, and the lotus, exists. According to this myth, a lotus flower gives birth to the god *Ra*[31], the most prominent deity in Egyptian cosmology. In this narrative, the lotus flower, opening and closing with the sunrise and sunset, is associated with the sun, symbolizing its cyclical nature. Another Egyptian deity associated with this symbol is *Nefertum*[32], whose name translates to "water lily." According to myths, both Nefertum and *Amun*[33] were born from the blue lotus flower. (Bolkhari, 2005, p. 105)

The trace of the use of the lotus motif during the Safavid era, or the era of Shah Abbas's flower, is evident with the influx of Chinese goods. However, through conducted studies, it can be observed that the primary inspiration for this motif can be traced back to *Manichaean*[34] art.

Based on a comparative analysis of images containing the lotus motif from the pre-Islamic and post-Islamic eras, it can be concluded that lotus motifs from the pre-Islamic era until the *Achaemenid*[35] period were mostly based on Egyptian patterns. These lotus motifs gradually moved towards Eastern motifs from the Parthian period due to the importance of Manichaean art influenced by Buddhist traditions. During the Islamic period, it can be observed that the lotus motif, especially the blue lotus, fully adheres to Eastern motifs influenced by Manichaean art. In the Timurid and Safavid periods, this motif took a more complete form, and the blue lotus motif

30 - Ogdoad of Hermopolis (Khmunu)
31 - He was believed to have ruled as the first pharaoh of Ancient Egypt. He was the god of the sun, order, kings, and the sky.
32 - Nefertem was, in Egyptian mythology, originally a lotus flower at the creation of the world, which had arisen from the primal waters.
33 - The god of wind.
34 - Manichaeism is a formerly major world religion, founded in the 3rd century CE by the Parthian prophet Mani, in the Sasanian Empire.
35 - The Achaemenid Empire or Achaemenian Empire, also known as the First Persian Empire was the ancient Iranian empire founded by Cyrus the Great of the Achaemenid dynasty in 550 BC.

in these periods can be considered one of the most beautiful and significant knotwork motifs. (Wilson, 1998, p. 138)



Figure No.14 – Examples of water lily patterns in ancient Iran and the Islamic period, (Lotfi, 2010), enhanced by the Author

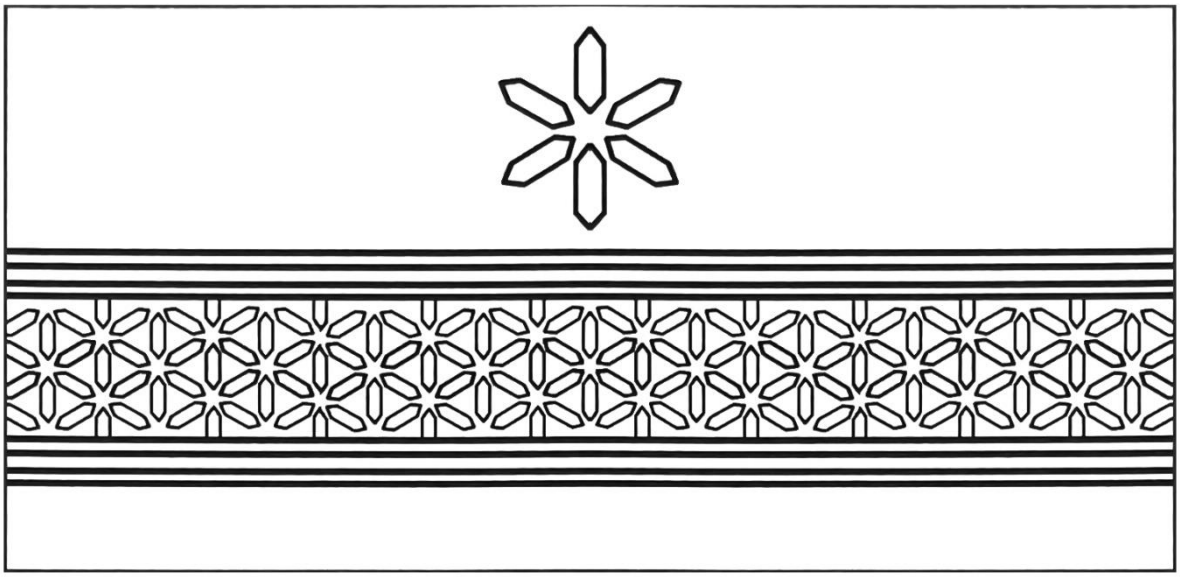


Figure No.15 – Example of repetitive and continuous pattern using the lotus flower motif (six-petal flower), gypsum decorations, Jameh Mosque of Isfahan, (reconstructed pattern), (Lotfi, 2010), enhanced by the Author

Geometric patterns in brickworks, tilework, and decorations

In the realm of brickwork decorations, the Safavids can be considered as the heirs of artistic traditions from the Timurid period. During this period, the brick coverings of domes reached their pinnacle of flourishing and evolution. Many structures from this era, including highly advanced caravanserais and the framework of Safavid-era buildings, utilized brickwork techniques. Furthermore, in the ornamentation of architectural elements from the Safavid period, such as the coverings of domes, ceilings of *Shabestans*[36], and other decorations, the art of tilework and mosaic work is quite prominent. It is worth noting that many of the motifs in the decorations of Seljuk architecture are also found in the art of brickwork and tilework during the Safavid period.

Among the brickwork techniques employed in the covering of domes, several notable methods include ribbed domes, conical domes, the pier-and-panel technique, four-section domes, four-pendant domes, and four-arched and four-ribbed domes. One of the most significant techniques is the Muqarnas[37] dome, a geometrically intricate design created using the principles of sacred geometry and the technique of subdivision.

The construction of many domes also utilized the double-shell technique, which was a prevalent method during the Safavid period. In this technique, two concentric layers of bricks are arranged in a dome-like structure with an empty space or gap between them. The inner layer serves as the structural support and framework, while the outer layer functions as a decorative element. Another brickwork technique extensively employed in the Safavid era, especially in tilework and

36 - A shabestan or shabistan is an underground space that can be usually found in the traditional architecture of mosques, houses, and schools in ancient Iran.
37 - Muqarnas, also known in Iberian architecture as Mocárabe, is a form of ornamented vaulting in Islamic architecture.

mosaic arrangements, is the zigzag connection. This method, aside from creating a robust bond between bricks, also serves as a decorative element in its own right. (Hillenbrand, 1994, p. 32)

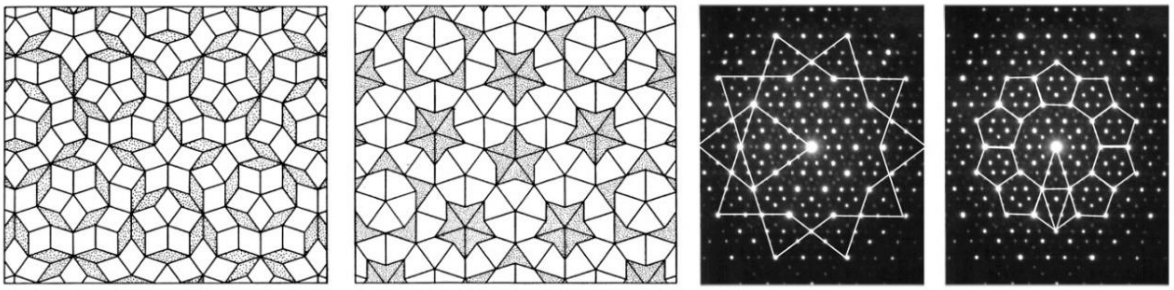


Figure No.16 – examples of tilework geometry. (Safavid and Zand periods), (reconstructed pattern), (url1), enhanced by the Author

Muqarnas, especially suspended Muqarnas, constitute one of the most crucial decorative elements, combining both brickwork and tilework techniques based on knotting. These are employed in the embellishments of domes, *minarets*[38], *mihrabs*[39], column capitals, the ceilings of Shabestans, and the inner walls of domes.

The pattern of Muqarnas in domes is formed by the intersection of square units that rotate around a specific axis, creating a star-shaped motif. This star-shaped form is among the highly significant patterns in Islamic designs. (Sakkal, 1982, pp. 20-21)

In many domes, Muqarnas, aside from their decorative role, have been utilized to transfer forces from the main bases (outer square) to the dome. (Upham, 1969, p. 102)

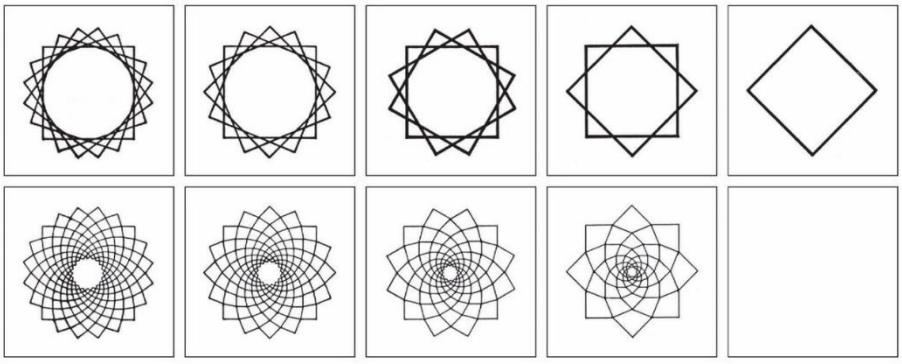


Figure No.17 – An example of geometric interlace patterns (from simple to complex), (reconstructed pattern), (Sakkal, 1989), enhanced by the Author

38 - A minaret is a type of tower typically built into or adjacent to mosques.
39 - Mihrab is a niche in the wall of a mosque that indicates the qibla, the direction of the Kaaba in Mecca towards which Muslims should face when praying.

Drawing Islamic geometric patterns and motifs with Sacred Geometry

In drawing geometric patterns in traditional arts, a compass and ruler are considered the two fundamental tools. This method is rooted in the technique of stretching a cord, which was historically used for designing buildings instead of compasses. (Lotfi, 2010, p. 154)

In the past, humans based their geometric drawings on circles, which is a complete method for shaping forms. In each circle, it is easy to create a regular polygon; you just need to divide the circumference of the circle into the desired number of sides by placing a point and then connecting adjacent points with equal straight lines to form the desired regular polygon. In this regard, let's review some of the features of drawing based on circles to facilitate the path for drawing more complex patterns, as the foundation for drawing Islamic and floral patterns is based on these few essential features.

To perform this drawing, according to Table 3 (first step), on line AB, a circular arc with an arbitrary radius (equal to the length of line AB) is drawn using a compass. (The unit size is taken as 1 in this context.) Then, from the center of the circle, which is the same as line AB, the diameter is divided into two equal parts. The two resulting equal line segments are the diameters of the small circles enclosed within the initial circle.

As a result of this drawing, as seen in Table 3 (second step), the well-known geometric pattern called "*Paisley*[40]" emerges. Boteh can be considered a geometric form similar to the Yin and Yang symbols, which play a significant role in most drawings, especially in the creation of Islamic patterns.

40 - Paisley or paisley pattern is an ornamental textile design using the boteh or buta, a teardrop-shaped motif with a curved upper end.

One of the characteristics of drawing such patterns is their fractal nature. If we repeat this process for smaller circles, a geometric escalation occurs. This escalation is such that if we consider the radius of the unit circle (R=1), the radii of the smaller circles successively become $\frac{R}{2}$, $\frac{R}{4}$, and so on, forming a mathematical escalation. (Al-Buzjani, 2005, p. 130) This escalation can be observed in Table 3 (third step).

Now, by maintaining the tangency of the smaller circles while having unequal radii, the same geometric pattern emerges with different proportions. We can observe this in Table 3 (fourth step). This new geometric pattern, with unequal proportions, closely resembles a leaf shape that serves as inspiration for Islamic patterns and floral patterns, playing a significant role in drawing these patterns.

The geometric form of a leaf is obtained by drawing circles based on a different principle. If these two circles have equal radii but are no longer tangent to each other and instead intersect at points c and d, the common space between them, indicated by a specific color, represents the desired leaf shape. This geometric form is of great importance in a particular case. In this scenario, if two circles with equal radii intersect each other at their centers (meaning if we connect the centers of the two circles, the length of the chord obtained becomes equal to the radius of the circles, precisely perpendicular to the chord cd), the common space obtained from the intersection of these two circles is the same as the Vesica Piscis shape that we observed in the theoretical section and the introduction of the sacred geometry pattern in Table 2.

The importance of this geometric pattern lies in its generative nature, similar to a fractal-generating seed, where all regular polygons can be drawn from its core. (Lawlor, 1989, p. 68) If we consider the radii of the circles as the unit size (r=1), the length of the line obtained from

connecting the two central points (cd) is equal to $\sqrt{3}$. (Lotfi, 2010) It is important to note that, in this research, given the lack of mathematical proofs, we refrain from proving this proportion. According to the Pythagorean theorem, an equilateral triangle can be drawn using the Vesica Piscis proportion, as shown in Table 3 (seventh step).

Moreover, if we enclose the mentioned pattern with a rectangle, the ratio of the sides of the surrounding rectangle becomes $\sqrt{3}$. This particular rectangle is called Hambidge[41]'s rectangle or the dynamic rectangle, and its significance lies in its proportions being based on the golden ratios found in nature. It possesses fractal properties and is reproducible, allowing the drawing of regular polygons using it.

For instance, to draw a regular hexagon, having the Vessica pattern, we draw an arc from one of the intersection points of two circles (for example, point c) with the same unit radius. If we extend this arc, precisely the same geometric pattern of three circles (Triquetra) emerges.

This arc from the second circle intersects the second circle at a new point, which we'll call e. Now, if we precisely repeat this process from point d, the new arc intersects the second circle at point g. We continue this process of drawing arcs from the new points e and g. (As observed, there is a fractal pattern and similar repetitions even in the process of drawing these geometric patterns.)

The two new arcs intersect each other on the circumference of the second circle at a point we'll call f. Now, if we connect all the obtained points (bcefgd), the desired regular hexagon is drawn. (Lawlor, 1989, p. 66)

41 - Jay Hambidge was an American artist who formulated the theory of "dynamic symmetry", a system defining compositional rules, which was adopted by several notable American and Canadian artists in the early 20th century.

The regular hexagon itself is formed by six equilateral triangles. By connecting the vertices of this hexagon to its center, a rectangle is created inside, as indicated in Table 3. Following the same table, we can draw the mentioned dynamic rectangle or the Platonic rectangle inside the hexagon. In ancient Western art, the ratio $\sqrt{3}$ was used by many artists even before other golden ratios. (Lotfi, 2010)

Now, if we draw the diagonals of this regular hexagon (which itself results in drawing the Platonic rectangle), we observe a crucial star-shaped pattern known as the hexagram, Star of David, or *Seal of Solomon*[42]. This pattern is one of the most important templates for drawing Islamic patterns, especially in tile designs.

Similarly, we can inscribe various regular polygons inside the circle, according to the third table. By utilizing computer-aided drawing, we can observe that the circumference of the circle is evenly divided by these polygons of different sides.

For example, another way to draw the shape of an equilateral triangle inscribed in a circle, using one of the abstract patterns of sacred geometry (using the Vesica Piscis pattern), is as follows: Firstly, the main diameter of the circle is drawn. Then, from the intersection point of the diameter with the circumference (point O), another circle with the radius of the original circle is drawn. As a result, the Vesica Piscis pattern is obtained. The two circles intersect at points A and B. Now, if we connect points A and B to each other and the starting point of the diameter, we form an equilateral triangle. (Al-Buzjani, 2005, p. 29)

42 - The Seal of Solomon or Ring of Solomon is the legendary signet ring attributed to the Israelite king Solomon in medieval mystical traditions, from which it developed in parallel within Jewish mysticism, Islamic mysticism and Western occultism.

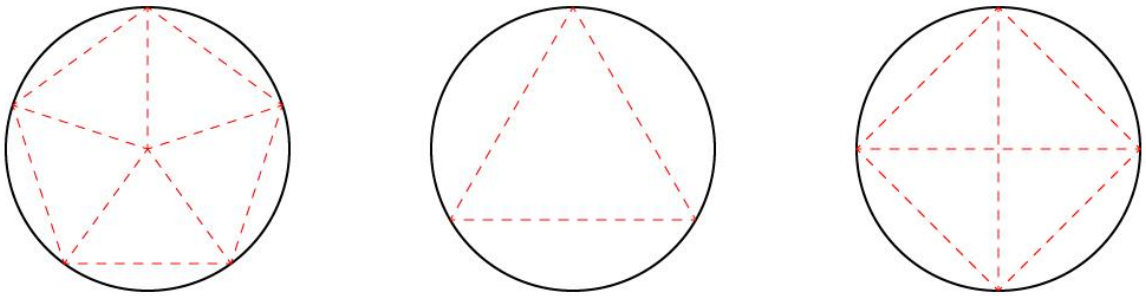
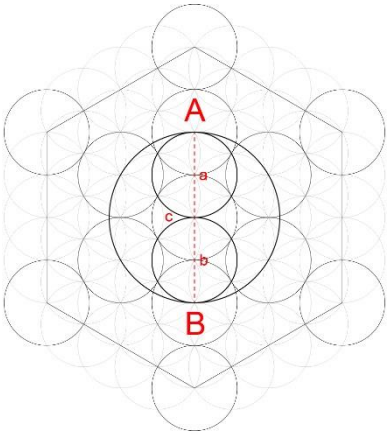
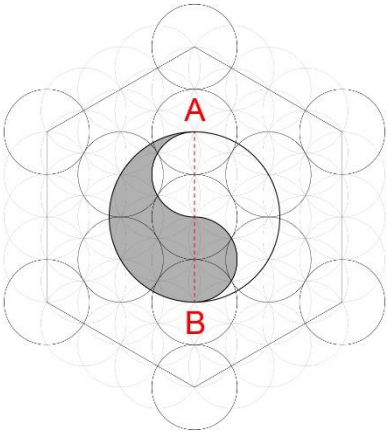
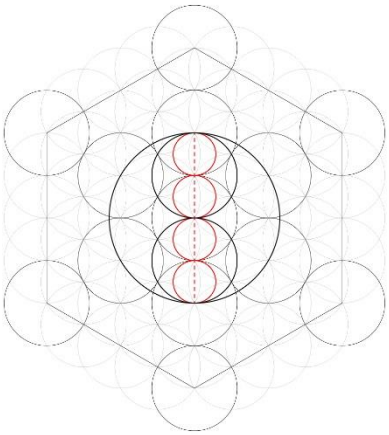


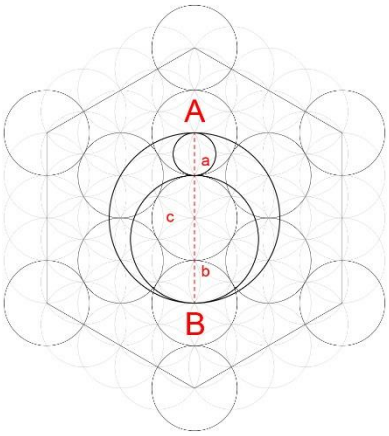
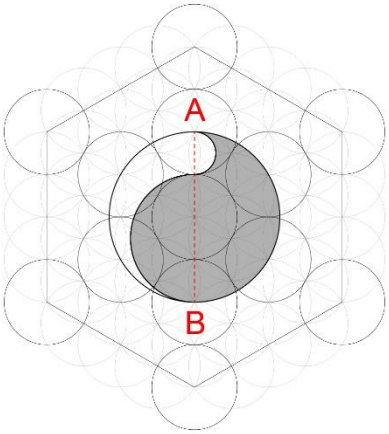
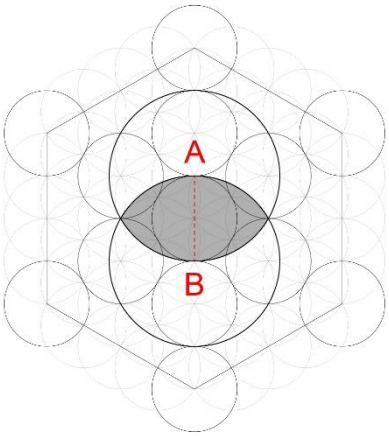
Figure No.18 – Examples of regular polygons inscribed in a circle (basic patterns for drawing Islamic geometries), done by the Author

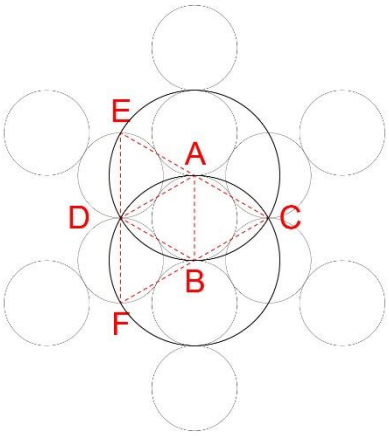
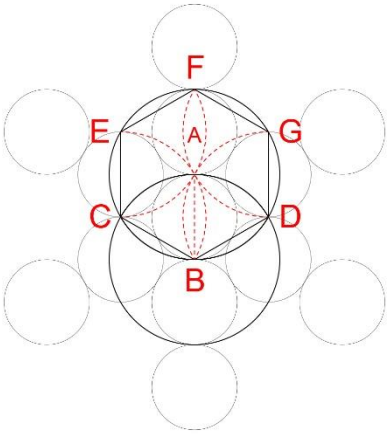
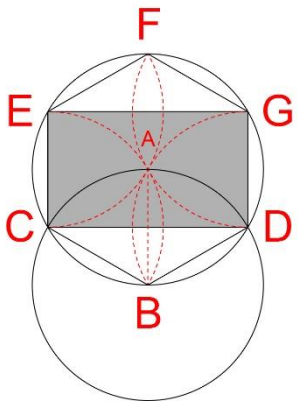
Now, to draw an inscribed square in a circle according to the third table, we first draw the vertical and horizontal diameters of the circle. Connecting the points where the diameters intersect the circumference creates a square inscribed in the circle. Now, by halving the sides of this square, we obtain points. Connecting these points to each other and extending them to the radius of the circle results in points A, B, C, and D.

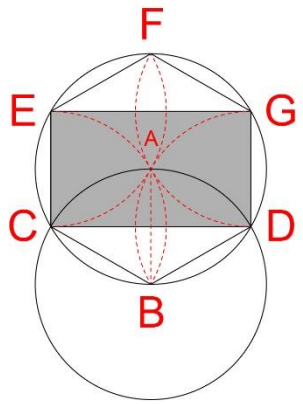
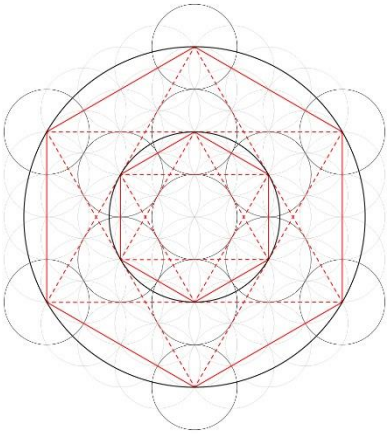
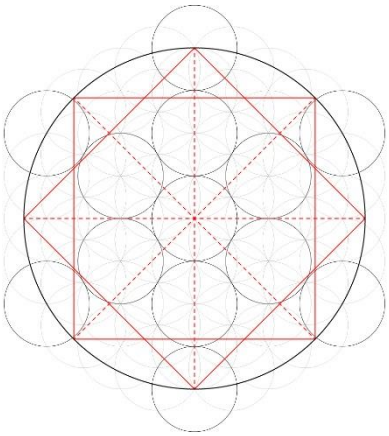
Connecting these points creates a second square inscribed in the circle. Combining these two squares also forms a 16-pointed star, one of the most familiar geometric patterns in brickwork and tilework, especially from the Safavid period onwards. (Al-Buzjani, 2005, p. 29) In general, the main structure of Islamic geometric patterns is created by using these patterns or combining them with different proportions.

Table no.3 – Drawing Islamic geometric patterns and motifs with Sacred Geometry, Drawing by the Author

Figure Name	comments	Figures
First Step	- Diameter = AB - ac = ab (length)	
Second Step	- Paisley pattern - Ying & Yang - Used in Islamic geometries.	
Third Step	- - forming a mathematical escalation. R/2, R/4, and so on. - Fractal Behavior	

Fourth Step	- Used in drawing of Islamic geometries and Floral motifs.	
Fifth Step	- the floral (leaf) pattern	
Sixth Step	- two overlapping circles - Vesica Piscis	

Seventh Step	- If we consider the radius of the circles to be a unit size (r=1), the line that results from connecting the two central points (cd) has a length equal to $\sqrt{3}$ - equilateral triangle	
Eighth Step	- regular hexagon - Sacred Geometry pattern	
Dynamic Rectangle	- Platonic rectangle - Golden Ratio	

Dynamic Rectangle	- Platonic rectangle - Golden Ratio	
Seal of Solomon	- Platonic rectangle - Golden Ratio	
Shamsa [16-pointed star]	- Main structure of Islamic geometric patterns.	

Geometrical studies of the dome of Sheikh Lotfollah Mosque

In this dome, there is a complete combination of the arts of geometry, tessellation, tilework, brickwork, and Islamic and Kufic patterns. Now that we have examined various examples in different periods of Iranian art and Islamic art, in this chapter, we will focus on the analysis and study of how the dome of the Sheikh Lotfollah Mosque is designed, the knotting techniques used, and the patterns adorning it.

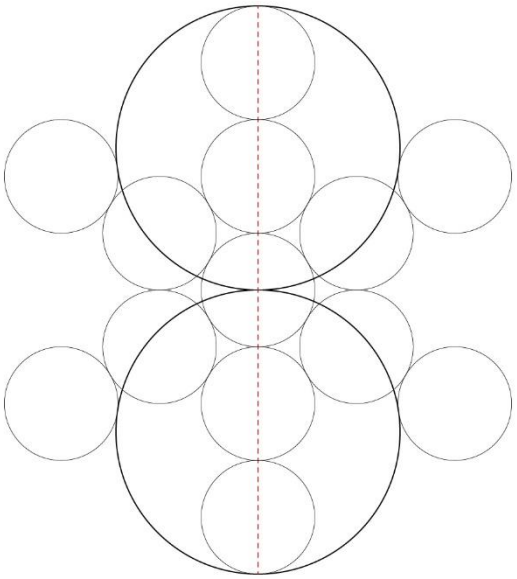


Figure No.19 – Tangent Circles, Drawing by the Author

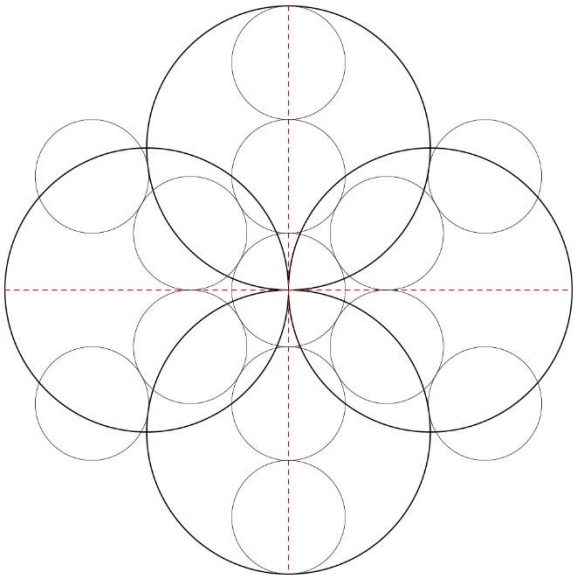


Figure No.20 – Pattern of Four Circles (with the repetition of the previous pattern), Drawing by the Author

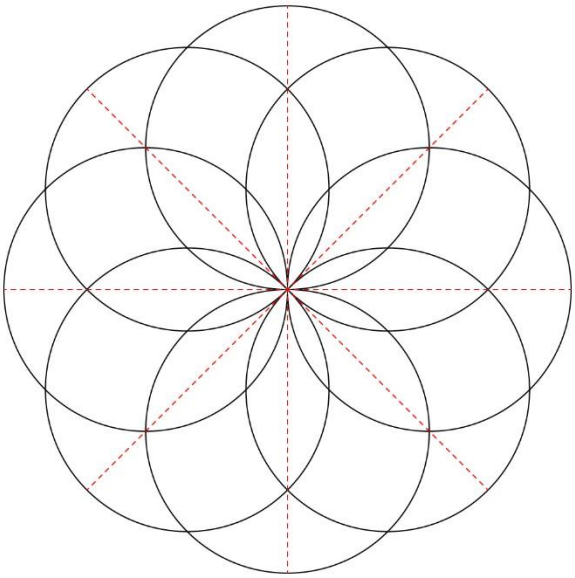


Figure No.21 – Eight circles, Drawing by the Author

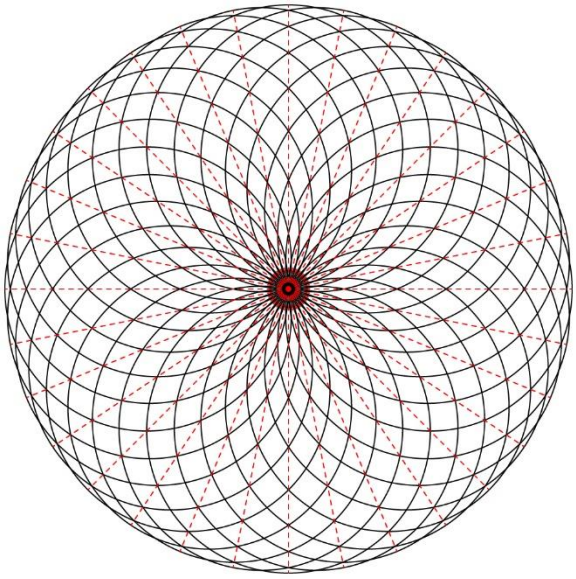


Figure No.22 – 32 circles, Drawing by the Author

As we can see, after drawing the initial pattern, by adhering to the principle of continuity and repetition, we arrive at the main pattern that forms the basis of the dome's decorations.

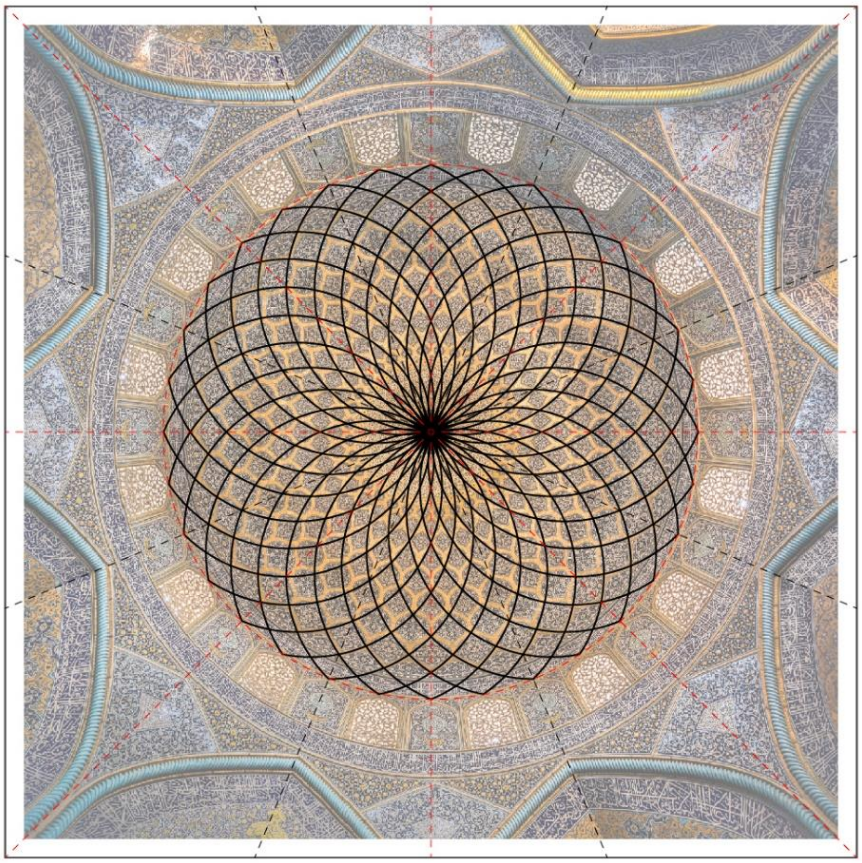


Figure No.23 – Knots and Geometric Patterns of Dome, Drawing by the Author

Now, onto this primary geometry, patterns are gradually added step by step. The closer elements are to the center (origin), the smaller and denser they are, and the farther they are from the center, the larger they become based on a specific mathematical proportion. This proportion and variation in size can be observed in the philosophy of abundance in unity and the perspective of the creatures' selflessness in relation to the Creator, as mentioned in the philosophical section.

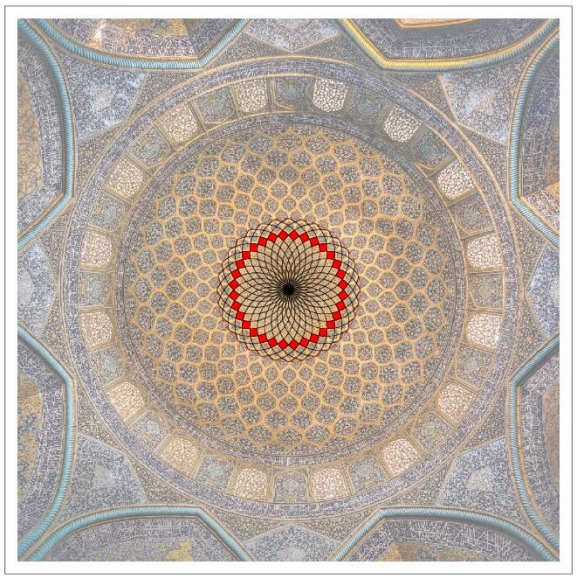


Figure No.24 – First layer, Drawing by the Author

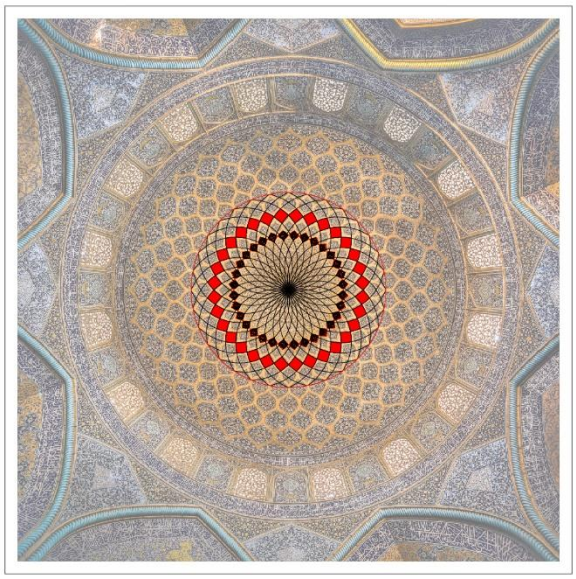


Figure No.25 – Second layer, Drawing by the Author

If this geometry is drawn step by step, we obtain the main nodes and the fundamental structure for placing the dome patterns. Now, with this structure, we can draw Islamic and Kufic patterns on the created framework and decorate the spaces between them with brickwork, patterned domes, and calligraphic inscriptions.

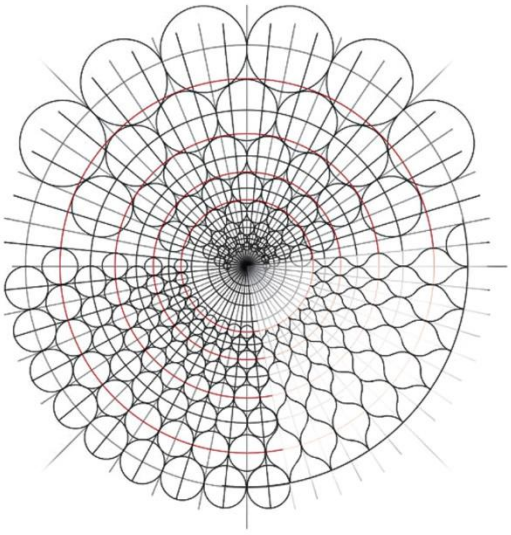


Figure No.26 – Main Structure of Dome Patterns (url1), Edited by the Author

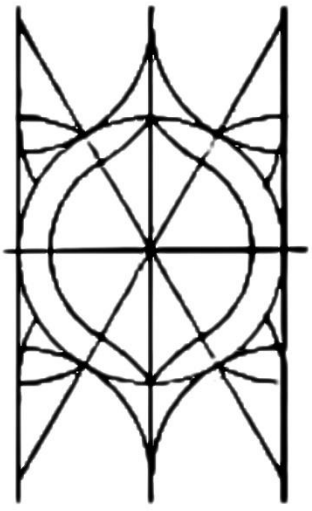


Figure No.27 – Pattern Used in Each Node (based on Vesica Piscis), (url1), Edited by the Author

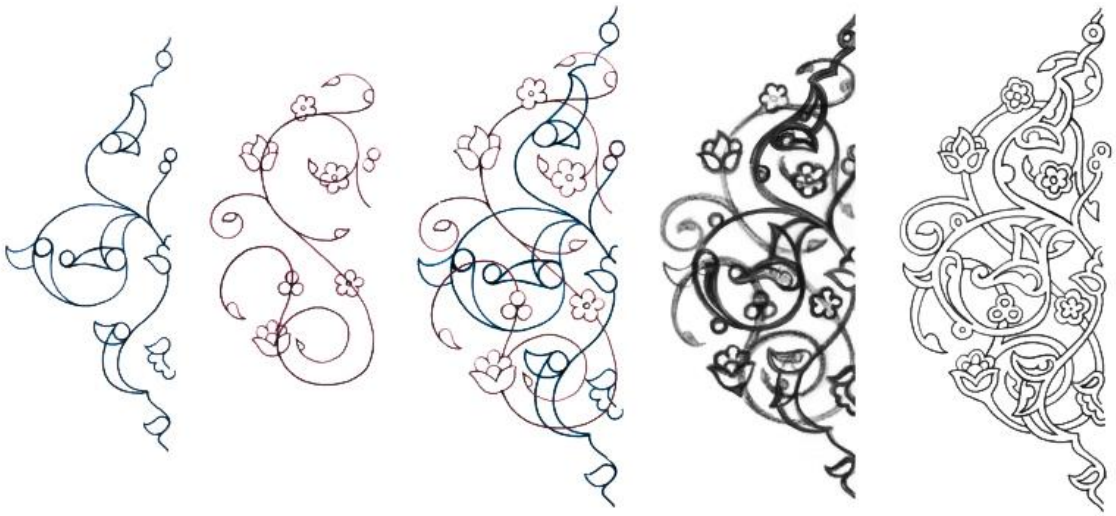


Figure No.28 – An Example of the Design Process of an Islamic Pattern on the Dome, (url1), Edited by the Author



Figure No.29 – Placement of Islamic and Khatayi (floral) Patterns in Each Section and the Use of Brickwork in the Border, (url3), Edited by the Author

We can consider the Vesica Piscis pattern as a kind of mother pattern in the initial drawings. As the patterns become more complex, we can observe the Tree of Life pattern in the structure of the designs, which is the geometric foundation for intricate patterns in both Islamic art and ancient world art.

Conclusion

In this research, an attempt was made to initially introduce abstract patterns of sacred geometry, both in theoretical sciences and in philosophical and historical sciences.

Platonic solids can be considered the first geometric patterns from which most geometric forms have, in a way, taken their shape from them. This signifies the central role of all geometric patterns after Plato emanating from sacred geometry. The adoption of these patterns from each other in an evolutionary manner in different periods, in a way that these structures, like other structures in nature, are constantly evolving, is indicative of this meaningful connection.

To investigate the second question, it can be concluded, based on geometric analyses and considering how Islamic and geometric patterns are drawn in tilework, that, similar to the first conclusion, the tools for drawing these patterns are the same compass and ruler used for sacred geometry. Apart from the consistent drawing tools, based on analytical diagrams, it is observed that the patterns for drawing these designs are based on circular patterns of sacred geometry, which themselves follow the golden ratios derived from nature.

In conclusion, as mentioned in the philosophical and theoretical sections, it can be noted that these geometric patterns have meaningful connections in the human mind and are inspired by their counterparts in nature. Due to their familiar motifs found in nature, these patterns consciously or subconsciously influence human perception. When we observe these patterns, we consciously or subconsciously recall the fundamental forms that make them up because these patterns are the first images we seem to have encountered in the world around us. As previously mentioned, Suhrawardi also talks about the connection with the imaginative world and considers these geometric patterns as a means of connecting with the imaginative world. The imaginative world

can be considered, based on the teachings of Eastern philosophy, as the origin of creation, a place where the soul is born. From this perspective, the experience of these patterns can bring divine meaning, especially in spatial designs, to life for us.

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